

Mridula Pillai

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Education

B.Tech Electronics and Communication Engineering with minor in B.Sc Physics

Aug 2021 – May 2025

Shiv Nadar Institution of Eminence, NCR

CGPA: 7.98

Courses: Antenna and Wave Propagation, Radar Engineering, Optical Engineering, Control Systems, Signal Processing, FPGA, VLSI Design, Embedded Systems, Digital Electronics.

Computational Exposure through MATLAB-based semiconductor modeling, FPGA interfacing, signal analysis, and python programming for scientific applications.

Internship

Square Kilometer Array (SKA) India Training Program, IIT Indore

Jul 2025 – Aug 2025

Organised by IITI, IITK, IIST, IUCAA, BITS and MAHE

- Designed and assembled a functional radio telescope, gaining hands-on experience in RF antenna design, impedance matching, feed network implementation, and low-noise signal reception.
- Worked on RF front-end hardware including LNA integration, filtering, coaxial interconnections, and signal conditioning with observational radio data and basic signal analysis workflows during practical astronomy sessions.
- Conducted practical training in radio astronomy, collaborated with researchers and peers from multiple institutes, analysing and interpreting astronomical signals using modern tools.

ABB India Limited, Bengaluru

Jan 2025 – May 2025

Project Intern - Process Automation Energy Industries

- Contributed to two automation projects using ABB System 800xA and AC800M, supporting requirement analysis, control logic development etc. across engineering and commissioning stages.
- Configured controllers, I/O modules, and field instrumentation, integrating plant-floor signals into the DCS via PROFIBUS and Modbus protocols.
- Assisted in testing and commissioning of automation systems in live factory environments, collaborating with cross-functional engineering teams.

Laboratory of Electro-Optics Systems (LEOS), ISRO, Bengaluru

May 2024 – Jul 2024

Summer Intern - Laser and Sensors Department

- Developed an FPGA–MATLAB GUI–based hardware control system for Erbium-Doped Fiber Amplifiers (EDFAs), integrating digital logic with an optoelectronic subsystem in an optical communication setup.
- Designed and implemented RTL and FSM-based control architecture in VHDL for automated parameter sequencing, UART-based communication, and real-time hardware interfacing, enabling deterministic control and system-level validation.
- Performed system-level debugging and validation by monitoring gain response under varying bias and operating conditions, correlating digital control signals with analog/optoelectronic device behavior.

Projects

Major Project I: Analysis of Photoconductive Antenna for Terahertz Generation

- Designed and analyzed photoconductive antennas with rectangular and interdigitated geometries for THz RF emission, evaluating transient carrier dynamics, radiation efficiency, radiated power, and impedance-related behavior.
- Developed physics-based models incorporating electron-hole transport, ultrafast excitation, and electric field distributions to characterize high-frequency current transients driving RF emission.
- Optimized electrode geometry, bias conditions, and semiconductor parameters to enhance radiated power, bandwidth, impedance characteristics, and far-field radiation performance, validated through MATLAB-based simulations.

VLSI Layout: 6T and 8T SRAM

- Designed and analyzed 6T and 8T SRAM cells, validating functionality, static noise margin (SNM), and robustness across PVT corners.
- Evaluated design trade-offs between high-speed/low-power 6T architectures and improved read stability in 8T cells, analyzing power consumption and reliability implications.

Memory Design & Layout on Cadence Virtuoso

- Designed and simulated 4×4 DRAM and SRAM arrays in Cadence Virtuoso, incorporating peripheral circuitry like row/column decoders, sense components etc. to validate memory functionality.
- Validated read, write, and hold operations through transient analysis, ensuring robust memory operation across multiple scenarios.
- Modelled process variations by tuning MOSFET channel lengths and evaluated performance under uniform and anti-correlated N/P conditions, exploring performance trade-offs under process variability.

Skills

- **Programming & Data Analysis:** Python, MATLAB, C, VHDL, Verilog.
- **Embedded & FPGA:** STM32, Arduino, Spartan-3.
- **Scientific / Simulation Tools:** Cadence Virtuoso, CST Studio, Xilinx ISE.

Interests

- Scientific Computing and Computational Modeling: For astronomical instrumentation.
- Radar, Antennas and Satellite Communication: 5G & 6G applications, Deep space communication and Astronomy applications.
- Observational Astronomy: Radio Astronomy.
- Semiconductor Architecture, SoC & FPGA Systems, Digital Logic Optimization.

Technical Activities & Outreach

Celestia Explora, Chapter of Students for the Exploration and Development of Space (SEDS)

Design and marketing head

International Association of Physics Students (IAPS)

Content and Design Team Lead